

Spatial Statistics - Homework 4.3

Pranav Tikkawar

March 5, 2026

1. 5.7

Derive 5.18:

$$\frac{\partial}{\partial \theta_j} \log |\Sigma(\theta)| = \text{tr}\{\Sigma^{-1}\Sigma_j(\theta)\}$$

Solution: Consider an m dimensional Gaussian random vector $Y \sim N(\mu(\beta), \Sigma(\theta))$.

Using the Jacobi identity $\partial(\det A) = \det A \cdot \text{tr}(A^{-1}\partial A)$, we have

$$\begin{aligned} \frac{\partial}{\partial \theta_j} \log |\Sigma(\theta)| &= \frac{1}{|\Sigma(\theta)|} \cdot \frac{\partial}{\partial \theta_j} |\Sigma(\theta)| \\ &= \frac{1}{|\Sigma(\theta)|} \cdot |\Sigma(\theta)| \cdot \text{tr}\{\Sigma^{-1}\Sigma_j(\theta)\} \\ &= \text{tr}\{\Sigma^{-1}\Sigma_j(\theta)\} \end{aligned}$$

2. 6.5 a

Solution: For a log-log plot, the slope of the line is given by the exponent of the power law. For all $m = 100, 200, 400$ we see that the slope is approximately .5 and when regraphing on a regular plot, we see that the line is approximately $y = \sqrt{x}$, which confirms our observation from the log-log plot.

3. 6.5 b

Solution: Let us consider the Taylor expansion of the function $K_\theta(x) = \theta_1 e^{-r/\theta_2} (1 + r/\theta_2)$ around $r = 0$:

$$\begin{aligned} K_\theta(x) &= \theta_1 e^{-r/\theta_2} (1 + r/\theta_2) \\ &= \theta_1 (1 - r/\theta_2 + (r/\theta_2)^2/2 - o((r/\theta_2)^2)) (1 + r/\theta_2) \\ &= \theta_1 (1 + r/\theta_2 - r/\theta_2 + (r/\theta_2)^2/2 - o((r/\theta_2)^2)) \\ &= \theta_1 (1 + (r/\theta_2)^2/2 - o((r/\theta_2)^2)) \end{aligned}$$

Notice that our principle irregular term is of order r^2 , with the coefficient $\theta_1/(2\theta_2^2)$.

Going back to the plot of θ_1 vs θ_2 , we see that the line of best fit on the log log scale

has a slope of approximately $\frac{1}{2}$ and that means the model that is fit is

$$\log \theta_2 = \frac{1}{2} \log \theta_1 + b$$

$$\theta_2 = B \cdot \theta_1^{1/2}$$

$$\frac{\theta_2}{\theta_1^{1/2}} = B$$

This finding is consistent with the fact that the coefficient of the principle irregular term is $\theta_1/(2\theta_2^2)$, which is constant when $\theta_2/\theta_1^{1/2}$ is constant.

Thus our reparameterization is given by $\phi_1 = \theta_2/\theta_1^{1/2}$ and $\phi_2 = \theta_2\theta_1^{1/2}$.

Note that we could have easily also chosen $\phi_1 = \theta_1/\theta_2^2$ and $\phi_2 = \theta_1\theta_2^2$, which would have also worked.